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A palaeomagnetic study of lower Proterozoic granites from Western Kenya

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Summary. This paper reports a palaeomagnetic study of the lower Proterozoic granites, exposed in the western part of Kenya and dated at 2530 ± 50 Ma ($\lambda_{Rb} = 1.39 \times 10^{-11} yr^{-1}$). Ten sites out of 13 have yielded stable palaeomagnetic directions which are believed to be primary. Polished section and thermomagnetic analyses indicate titanomagnetite grains as the main carrier of the primary component of the natural remanence. The mean palaeomagnetic north pole is calculated at 210.7° E, 60.9° N with $A_{95} = 13.5°$.

1 Introduction

The African polar wander curve for the intervals 2700–2300 and 1970–1300 Ma was first defined by McElhinny *et al.* (1968) from the studies of the rocks of the Rhodesian–Kaapvaal cratons. During the last few years more palaeomagnetic results of the Precambrian rocks from other parts of Africa have been published. The palaeomagnetic data from Kenya (Brock, Raja & Vise 1972), Tanzania, Zambia and West Africa (Brock & Piper 1972; Piper, Briden & Lomax 1973), South Africa and south-west Africa (Piper 1975) have been used to define the polar wander curve for the interval 1300–500 Ma.

Piper *et al.* (1973) defined a single apparent polar wander swathe for the interval 2300–1950 Ma using the palaeomagnetic results from the Kaapvaal, Congo and West African cratons. They suggested that the principal cratonic areas of Africa have been approximately in the present relative positions and orientations since 2200 Ma. So far only six formations (listed in Table 3) older than 2300 Ma have been studied palaeomagnetically. Consequently, the polar wander curve is poorly defined for periods prior to 2300 Ma.

The granites in the western Kenya have been dated at 2530 ± 50 Ma by the whole-rock Rb–Sr isochron method, using $\lambda_{Rb} = 1.39 \times 10^{-11} yr^{-1}$ (Dodson *et al.* 1975). This Rb–Sr isochron age is in close agreement to the date of 2595 ± 100 Ma, a K–Ar determination on a hornblende by Cahen & Snelling (1966).

This palaeomagnetic study of the granites was undertaken with a view to add further information to the apparent polar wander curve for the range of 2700–2300 Ma. The palaeomagnetic results from this study are the first results from outside South Africa in the relevant age range.

2 Geology

The granites form the major intrusions in the Kavirondian and the Nyanzian Systems of the Western Province. The Nyanzian and the Kavirondian Systems are presumed to be the oldest Precambrian rocks in Kenya. The Kavirondian System unconformably overlies the Nyanzian System. The rocks of both the systems were folded after the deposition of the Kavirondian System (Cahen & Snelling 1966). These two systems have suffered thermal metamorphism due to the intrusions of the granites.

In general, the texture of the granites varies from fine to extremely coarse grained. The usual colour of the granites is mottled pink or white and black. The pink colour has been attributed either to hematite flakes in feldspar or to an iron-rich solution derived elsewhere (Pulfrey 1945). The coarse-grain variety carries closely packed large 'salmon' type feldspar phenocrysts. Quartz is generally interstitial and often occurs as granular aggregates with the feldspar. The dark minerals are usually hornblende and biotite. The accessories are apatite and sphene.

Pulfrey (1945) suggested the possibility that there is more than one intrusion in the Mumias granite, west of Broderick Falls (Fig. 1). His interpretation was based on the varied texture of the rock and considerable variations observed in volumetric modes of the rock in that area. On the other hand, no exposures of contacts of the fine- and the coarse-grained granites have been reported in that area.

3 Sampling and measurements

A total of 56 oriented block samples was collected from 13 sites which are shown in Fig. 1. A portable sampling drill could not be used due to the extreme hardness of the granites.

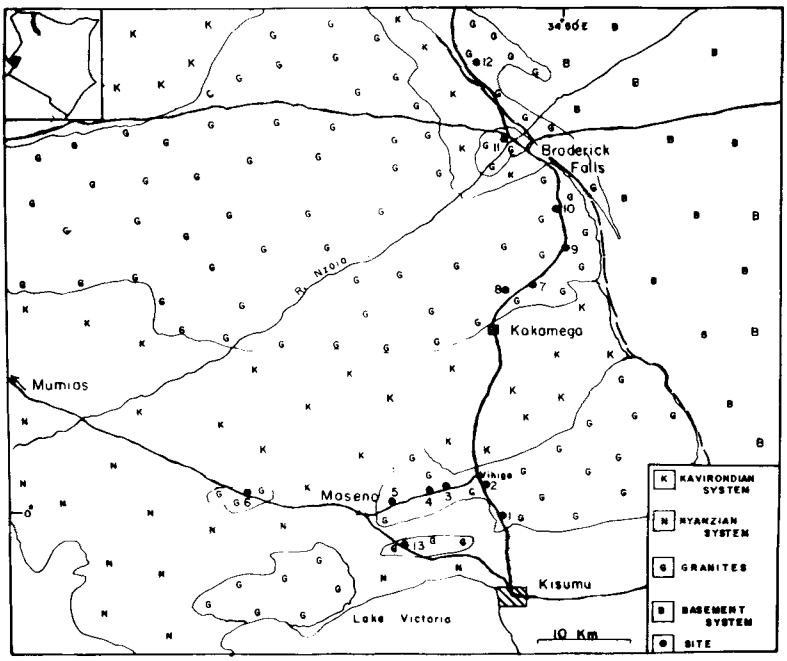


Figure 1. Simplified geological map illustrating the occurrence of the granites and the locations of the sampling sites in western Kenya.

The samples collected include Kisian granite (site 13), Maragoli granite (sites 1–5), Mumias granite (sites 7–12), and Nyarodi granite (site 6). Because of the heavy oxidizing conditions which exist in the Western Province of Kenya, good natural outcrops were difficult to find. Most of the samples were collected from freshly exposed rock at road cuttings. The Oyugis granite outcrops, between Kindu Bay and Oyugis (not shown in Fig. 1), south of Kismu township, were found to be typically boulderly and no samples were collected from this area.

Core samples of 2.2 cm diameter were taken from the blocks and specimens of 2.5 cm length were cut from the cores. Their NRMs were measured on a commercial version of the 5 Hz Foster (1966) type of spinner magnetometer. It was inferred from the af demagnetization curves of the pilot specimens that the rock possessed weak and inhomogeneous magnetization with low coercive force. Hence, all the granite specimens were subjected to stepwise af demagnetization in steps of 2.5 mT up to a maximum peak field of 40 mT except those from site 8, which were treated up to 70.0 mT peak field only.

The variations in within-site dispersion may be illustrated by plotting k/k-significant ($k$sig.) versus the af field as described by Irving, Stot & Ward (1961). The precision parameter k is given by $(N-1)/(N-R)$ where R is the resultant vector of a sample size N. The length of the vector will be large if a preferred direction exists and small if not. Assuming no preferred direction exists, a value R_0 may be calculated which will be exceeded by R with any stated probability. Hence, k/ksig. is equal to $(N-1)/(N-R_0)$. If $R < R_0$, then $k < k$sig. When the ratio k/ksig. is less than unity, the distribution may be considered random. The highest ratio of k/ksig. indicates the closest grouping of the specimens directions in a site. Consequently, the direction, when the ratio was observed to be the highest, was taken as the site mean direction in the present study. The site mean directions were then given unit weight to evaluate the mean direction of the granite formation.

4.1 RESULTS AND DISCUSSIONS

The palaeomagnetic results of the 13 sites are tabulated in Table 1. The site mean directions before and after af demagnetization are plotted on equal area stereo-nets in Fig. 2. The total NRM directions are observed to be scattered mainly in the upper hemisphere. The scatter of the directions was reduced on demagnetization as shown in Fig. 2(b). This can also be seen from the tabulated results (*cf.* Table 1) where the majority of the sites showed significant increase in the values of the precision parameter k after the af treatment. The results show that the formation has mixed polarity. The k/ksig. ratio for site 3 remained below unity at all fields of demagnetization. It was observed that the mean sample directions of this site remained scattered with individual specimens moving erratically in an unstable fashion. Consequently, this site was considered random and discarded from the final analysis. All the other sites are significant on the basis of the criterion used.

Site 6 gave a good 'reversed' direction and its polarity was reversed in the evaluation of the mean direction of the significant sites. Sites 8 and 12 apparently gave stable directions but their divergences from the group mean direction are extremely large. Therefore it is unlikely that they belong to the population sampled. Besides this, the natural remanence of the specimens from these two sites is suspected to be of a secondary origin (refer to Section 4.2). Therefore, these two sites, 8 and 12, were discarded from the final analysis on the basis of the above two reasons. Hence, 10 sites were used in the final analysis.

An overall mean direction ($D = 2.7°$, $I = -47.8°$, $\alpha_{95} = 13.8°$) has been calculated from the mean directions of the 10 sites (*cf.* Table 2). The palaeomagnetic pole is calculated by

Table 1. Site mean directions and poles of the granites.

		Total NRM				After A.F. cleaning						Paleomagnetic north pole			
Site Number	N	Dec. D	Inc. I	R	k	Dec. D	Inc. I	R	k	α_{95}	Highest Ratio k/ksig.	Haf mT	Long. (E)	Lat. (N)	dm	dp
1	5	26.6	-61.5	4.6	9.8	9.2	-54.9	4.9	52.8	10.6	19.6	20.0	203.9	53.4	15.0	10.6
2	4	333.6	-68.6	3.2	3.7	22.0	-76.8	3.9	48.0	13.4	14.5	12.5	205.3	23.1	24.9	23.2
3	4	178.3	-36.4	1.6	1.3	177.3	-12.4	2.7	2.3	79.9	0.7	r a n d o m				
4	4	15.4	-68.5	4.2	5.5	353.2	-47.9	4.0	99.2	9.3	30.0	25.0	227.3	60.2	12.1	7.9
5	5	349.7	-59.4	4.1	4.7	5.6	-44.9	4.9	35.1	13.1	13.0	7.5	204.3	62.9	16.6	10.5
6	3	250.4	- 1.6	2.0	2.1	184.3	+49.3	2.7	11.9	37.4	2.3	20.0	27.0	-59.5	49.6	32.9
7	4	41.5	- 8.9	3.7	10.0	41.5	-22.3	3.9	26.7	18.1	8.1	15.0	142.3	47.1	19.2	10.1
8	5	195.4	-31.8	4.6	12.5	200.2	-34.3	4.7	13.0	22.0	4.8	7.5	261.4	-62.9	25.2	14.4
9	4	330.5	-25.6	3.2	4.1	334.5	-39.2	3.5	6.2	39.9	2.0	25.0	261.1	56.4	47.7	28.5
10	4	3.6	- 7.9	2.9	2.9	354.9	-29.1	3.9	20.9	20.9	6.3	25.0	233.3	73.3	22.7	12.5
11	5	318.7	-31.2	4.6	9.2	348.8	-37.9	4.0	121.3	8.4	45.0	10.0	241.3	65.5	9.9	5.9
12	3	41.9	-42.0	2.7	5.9	39.8	+42.0	2.7	6.5	53.0	1.3	20.0	90.2	44.8	65.0	39.0
13	3	8.8	-30.3	2.8	9.2	6.8	-39.6	2.9	64.1	15.5	12.8	7.5	199.4	66.8	18.6	11.1

N, number of samples. *D*, declination, east of TN, degrees. *I*, inclination, positive down, degrees. *R*, resultant vector. *K*, *k*, Fisher's precision parameter. A_{95}, α_{95}, radius of 95 per cent circle of confidence (°). Haf, optimum cleaning field, mT. *dm*, *dp*, semi axes of 95 per cent oval of confidence (°). Long., Lat., longitude and latitude of palaeomagnetic north pole (°).

two methods. A single pole with an oval of confidence is obtained at 210.0° E, 60.7° N (dm = 18.0°, dp = 11.7°) from the mean of the site directions. The other mean pole, with a circle of confidence, is obtained at 210.7° E, 60.9° N (A_{95} = 13.5°) from the site poles listed in Table 2. The mean palaeomagnetic pole, computed from the site poles, was preferred since the sites were geographically widespread. The pole position of the granites, obtained from the present study, may be considered as that of the ancient geomagnetic pole for the

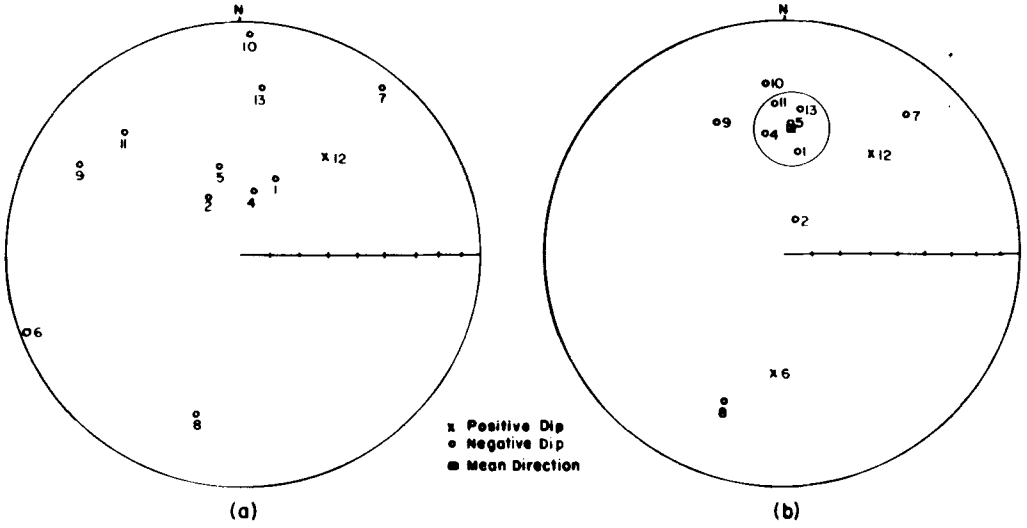


Figure 2. Site mean directions of the non-random sites: (a) before, (b) after af demagnetization.

Table 2. Formation mean direction and poles after af demagnetization.

	D	I	α_{95}	k	Long. (E)	Lat. (N)	K	dm	dp	A_{95}
Mean of the Poles of the 10 significant sites	-	-	-	-	210.7	60.9	13.9	-	-	13.5
Poles from the mean direction of the 10 significant sites	2.7	-47.8	13.8	13.3	210.0	60.7	-	18.0	11.7	-

following reasons. First, most of the mean directions are well removed from the direction of the present Earth's field in the area. (The direction of the Earth's field in the Kakamega area is about $D = 2^\circ$ W, $I = -22^\circ$, Skinner 1975.) Secondly, the improvement in grouping after cleaning both the sample and the site directions is a strong evidence of the primary direction of magnetization. Thirdly, the possibility of ancient secondary magnetization is ruled out since there is no evidence of reheating the granites since the rock was formed. The Pan-African thermal and orogenic event (450–700 Ma) has not so far been recorded in the areas where the granites were sampled (Cahen & Snelling 1966).

4.2 OPAQUE AND THIN SECTIONS PETROLOGY

Polished and thin sections of the specimens of all the sites were examined under the microscope. The polished section petrology showed that the specimens of all the sites contained titanomagnetite grains of size ranging from about 50–350 μm in diameter. In a few cases the grain boundaries were corroded and possessed inclusions of silicate minerals. High-temperature oxidation of the grains, class 2 type of Ade Hall *et al.*'s (1968) classification, was observed only in specimens of sites 5 and 11. Ilmenite lamellae were exsolved in the titanomagnetite grains.

Highly corroded titanomagnetite grains in the specimens of sites 8 and 12 were partly replaced by a hematite phase. The average Curie temperatures of the specimens of these two sites were obtained at 600 and 590°C, respectively, which confirmed the presence of hematite. The initial formation of the hematite phase appeared to be along the cracks and corroded margins of the host grains. This mode of occurrence of the hematite possibly suggested that it was non-deuteric. This secondary hematite may have acquired a CRM at a much later date after the solidification of the granites.

Thin sections were examined to find any difference in the degree of alterations of the silicate minerals. The feldspars, observed in all the specimens, were heavily clouded and siricitized. The main mafic minerals observed were hornblende, augite, biotite and hypersthene. Hornblende was often chloritized and associated with opaque granular grains, possibly magnetite. However, no significant differences in mineralogy were observed between the specimens of the significant and discarded sites.

4.3 AFRICAN POLAR WANDER PATH FOR THE INTERVAL 2700–2300 MA

Piper (1976) summarized the Precambrian results from Africa for the interval 2700–2300 Ma. Table 3 gives the list of poles which he has used in his discussions. Fig. 3 shows the pole path as suggested by Briden (1976) using Piper's (1976) results. The pole path suggested is similar to that of McElhinny *et al.* (1968).

Table 3. African Precambrian palaeomagnetic poles for the interval 2700–2300 Ma. The listed numbers of the poles are those used by Piper (1976).

Pole Number	Rock Unit	Age (Ma)	Mean Paleomagnetic Pole	A_{95}
1	Modipe gabbro, Botswana	2630 ± 470	33°N, 211°E	10.5
2	Great Dyke, Rhodesia	2520 ± 30	21.5°N, 61.5°E	9
3	Jeppestown Amygdaloidal lava, South Africa	2500	10°N, 103.5°E	-
4	Gaberones Granite, Botswana	2340 ± 50	34.5°N, 103.5°E	16
6	Lower Ventersdorp lavas, South Africa	2500	4°N, 220°E	11
7	Upper Ventersdorp lavas, South Africa	2300 ± 100	71°N, 173°E	9

Pole number 3 of the list (Table 3) is from only one site and pole number 6 has been probably remagnetized at about 2200 Ma (Briden 1976). Consequently these two poles, 3 and 6, have been omitted from the pole path (Fig. 3). Three of the poles (1, 4 and 7) are from the Transvaal craton and pole 2 is from the Rhodesian craton.

An attempt has been made to fit the granite pole (2530 ± 50 Ma) on the pole path suggested by Briden (1976). Poles G1 and G2, plotted on the Precambrian pole path in Fig. 3, are the north and south poles, respectively, of the granites. The results of the granites may be fitted on to the proposed path in two ways. If the north pole at 61° N, 211° E is considered, then it suggests that pole G1 is older than pole 1 (the Modipe gabbro pole). If it is the south pole at 61° S, 31° E, then it suggests that G2 is younger than pole 7, the one from the upper Ventersdorp lavas. However, the latter is unlikely, but the former is possible when the error ranges of poles G1 and 1 are taken into account. Pole G1 is the first pole outside South Africa in the interval under consideration and it is rather too early to say how it fits on the proposed swathe by Briden.

It has now been well established that the Rhodesian craton has not moved relative to the Transvaal craton since 2260 Ma (Briden 1976). If one accepts Piper's proposition that Africa has been unity for much of Precambrian time, his evidence for it does not go back further

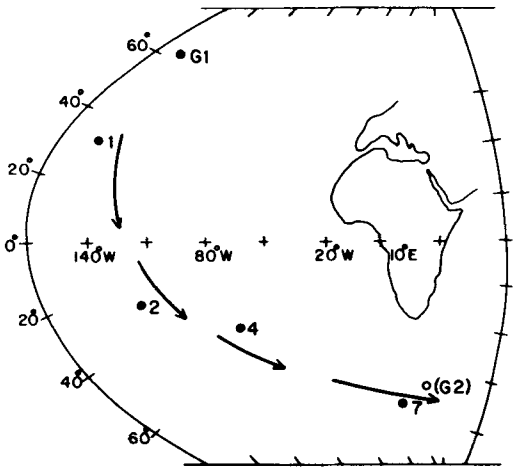


Figure 3. The plot of the granite poles – G1 North Pole, G2 South Pole. The other poles are those used by Piper (1976).

than about 2200 Ma. Pole G1 is from East Africa, a completely different cratonic block. It is quite possible that there may have been relative movement between them.

5 Conclusions

Pulfrey's (1945) suggestion of more than one intrusion of the granites is based on only the textural and the modal variations of the granites. However, the differences in the texture and the composition could be also due to variations within the body of one distinct mass. The granites may be differentiation facies of one unit rather than separate intrusions.

The age determinations of the Mumias granite from the two different locations are similar. The Buteba granite, possibly continuation of the Mumias granite into Uganda, has been dated at 2535 ± 125 Ma (Cahen & Snelling 1966) – a similar age to that of the Mumias granite. Consequently the age relationship of the Mumias and Buteba granites possibly rules out the chances of the intrusions of varied ages. On the other hand, the occurrence of the mixed polarity makes the interpretation more complicated. Whether there have been short reversals of the magnetic field of the Earth during the period 2700–2300 Ma is not yet known.

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